

Credit Risk Model - Performance Report

1. Project Overview

This project predicts whether a loan applicant is likely to default on their credit. The model was trained on the Credit Risk Dataset using a Random Forest Classifier with 100 estimators.

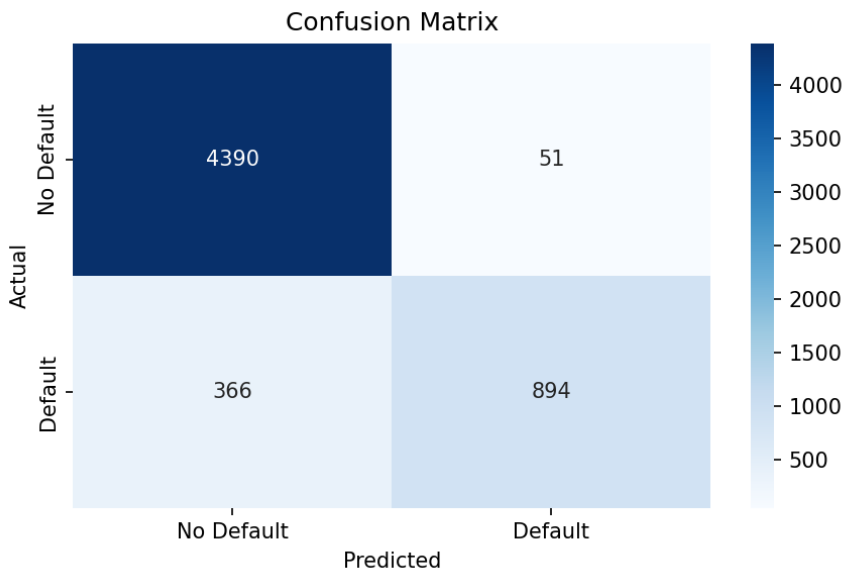
2. Model Used

Algorithm : Random Forest Classifier
n_estimators : 100
random_state : 42
Test Set Size : 5701 samples

3. Model Performance Metrics

Overall Accuracy : 92.69%
Precision (Class 1 - Default) : 94.60%
Recall (Class 1 - Default) : 70.95%
F1 Score (Class 1 - Default) : 81.09%

4. Confusion Matrix



5. Business Interpretation

The model achieves 93% overall accuracy. It correctly identifies 71% of actual defaulters (recall), which is critical for minimizing financial risk. With 95% precision on default prediction, false alarms are kept low. This model is reliable for use in automated loan screening systems.

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6. Deliverables

- model.ipynb : Full model training notebook
- cleaned_dataset.csv : Preprocessed dataset
- accuracy_report.pdf : This report